MILITARY SPECIFICATION

ANTENNAS, FIXED, HIGH FREQUENCY, GENERAL SPECIFICATION FOR

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 Scope. This specification covers the general requirements for fixed, shore-based, high frequency (HF) antenna systems for use at Naval Shore Communications Stations, hereinafter referred to as the antenna. This specification does not cover rotatable log periodic antennas or other antennas requiring mechanical movement. Specific details are specified in the applicable detail specification (see 3.1).

1.2 Classification. Antennas shall be of the powers and wind classes specified in 1.2.1 and 1.2.2, and as specified in the applicable detail specification (see 6.2).

1.2.1 Power input capability. Antennas shall be of the following powers as specified (see 6.2):

Power 0	250 watts average	1 kilowatt (kW) peak instantaneous (receive)
Power 1	1 kW average	4 kW peak instantaneous
Power 5	5 kW average	25 kW peak instantaneous
Power 10	10 kW average	40 kW peak instantaneous
Power 10a	10 kW average	80 kW peak instantaneous
Power 25	25 kW average	100 kW peak instantaneous
Power 50	50 kW average	400 kW peak instantaneous

1.2.2 Wind classes. Antennas shall be of the following wind classes, as specified (see 6.2):

Class	Wind and ice conditions
1	100 miles per hour (mph) (no ice)
2	125 mph (no ice) 90 mph with 0.5 inch (in.) radial ice on all members
3	150 mph (no ice) 90 mph with 0.5 in. radial ice on all members
4	Special case, as specified in the contract (see 6.2)

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbook. Unless otherwise specified, the following specifications, standards, and handbook of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

W-R-550	Rod, Ground (With Attachments)
QQ-B-654	Brazing Alloys, Silver
QQ-S-763	Steel Bars, Wire Shapes, And Forgings Corrosion-Resisting
RR-S-550	Sockets, Wire Rope
RR-S-1301	Safety Equipment, Climbing
TT-P-641	Primer, Coating, Zinc Dust-Zinc Oxide (For Galvanized Surfaces)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Electronic Systems Command (ELEX-8111), Washington, DC 20363, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

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MIL-I-10	Insulating Compound, Electrical, Ceramic, Class L
MIL-P-116	Preservation, Methods Of
MIL-C-3650	Connectors, Coaxial, Radio Frequency, Series LC
MIL-A-9094	Arrestor, Lightning, General Specification For
MIL-C-16173	Corrosion Preventive Compound, Solvent Cutback, Cold Application
MIL-E-16400	Electronic, Interior Communication And Navigation Equipment, Naval Ship And Shore: General Specification For
MIL-E-17555	Electronic And Electrical Equipment, Accessories, And Repair Parts, Packaging And Packing Of
MIL-F-17655	Field Changes And Field Change Kits; General Specification For
MIL-I-17918	Insulator, Strain, Rigging
MIL-P-21035	Paint, High Zinc And Dust Content, Galvanizing Repair
MIL-T-22361	Thread Compound Antiseize Zinc Dust-Petrolatum
MIL-T-28732	Transformer, Impedance Matching, Balanced To Unbalanced (BALUN)
MIL-P-28766	Poles, Structural Wooden, For High Frequency (HF) Antennas
MIL-T-28767	Transformer, Impedance Matching (Balun), Receive Only
MIL-S-38228	Sealing Compound, Environmental, For Aircraft Surfaces
MIL-C-39012	Connectors, Coaxial, Radio Frequency, General Specification For
MIL-I-45108	Inspection System Requirements
MIL-A-46106	Adhesive-Sealants, Silicone, RTV, General-Purpose (For Electrical And Mechanical Sealing)
MIL-C-83488	Coating Aluminum, Ion Vapor Deposited

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures And Tables For Inspection By Attributes
MIL-STD-108	Definitions Of And Basic Requirements For Enclosures For Electric And Electronic Equipment
MIL-STD-109	Quality Assurance Terms And Definitions
MIL-STD-454	Standard General Requirements For Electronic Equipment
MIL-STD-461	Electromagnetic Emission And Susceptibility Requirements For The Control Of Electromagnetic Interference
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement Of
MIL-STD-810	Environmental Test Methods And Engineering Guidelines
MIL-STD-965	Parts Control Program
MIL-STD-1364	Standard General Purpose Electronic Test Equipment

HANDBOOK

MILITARY

MIL-HDBK-419	Grounding, Bonding And Shielding For Electronic Equipments And Facilities

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2.1.2 Other Government publications. The following other Government publications form a part of this specification to the extent specified herein.

PUBLICATIONS

NAVAL FACILITIES ENGINEERING COMMAND (NAVFAC)

NAVFAC DM-2.2	Design Manual, Structural Engineering Loads
NAVFAC DM-7.2	Design Manual, Foundations And Earth Structures

FEDERAL AVIATION ADMINISTRATION (FAA)

FAA Advisory Circular 70/7460-1	Obstruction Marking And Lighting

(Copies of specifications, standards, handbooks, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN INSTITUTE OF STEEL CONSTRUCTION (AISC)

Manual Of Steel Construction (latest edition)

(Application for copies should be addressed to the American Institute of Steel Construction, Inc., 400 N Michigan Avenue, Chicago, IL 60611.)

THE ALUMINUM ASSOCIATION (AA)

Aluminum Construction Manual,
Aluminum Association Standard
"Specification For Aluminum Structures" (latest edition)

(Application for copies should be addressed to The Aluminum Association, 818 Connecticut Avenue, NW, Washington, DC 20006.)

AMERICAN WELDING SOCIETY (AWS)

AWS C2.2	Recommended Practice For Aluminum and Zinc For Protection Of Iron And Steel
AWS D1.1	Structural Welding Code

(Application for copies should be addressed to American Welding Society Inc., 550 Northwest Le Jeune Road, Miami, FL 33126.)

NATIONAL FOREST PRODUCTS ASSOCIATION (NFORP)

National Design Specifications For Stress-Grade Lumber And Its Fastenings

(Application for copies should be addressed to National Forest Products Association, Forest Industries Building, 1619 Massachusetts Avenue, NW, Washington, DC 20036.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI C29.1	Electrical Power Insulators, Test Methods
ANSI C29.2	Wet-Porcelain Insulators (Suspension Type)
ANSI C29.3	Wet-Porcelain Insulators (Spool Type)
ANSI C29.4	Wet-Process Porcelain Insulators (Strain Type)
ANSI C29.5 and C29.5a	Wet-Process Porcelain Insulators (Low- And Medium-Voltage Pin Type) Including Supplement
ANSI C29.6 and C29.6a	Wet-Process Porcelain Insulators High-Voltage Pin Type) Including Supplement
ANSI C29.7 and C29.7a	Wet-Process Porcelain Insulators (High-Voltage Line-Post Type) Including Supplement
ANSI C29.8	Wet-Process Porcelain Insulators (Apparatus, Cap And Pin Type)
ANSI C29.9	Wet-Process Porcelain Insulators (Apparatus, Post Type)

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A123	Zinc (Hot-Galvanized) Standard Specification For Coatings On Products Fabricated From Rolled, Pressed And Forged Steel Shapes, Plates, Bars And Strip
A153	Zinc Coating (Hot Dip) On Iron And Steel Hardware
A194	Nuts, Carbon And Alloy Steel, For Bolts For High Pressure And High-Temperature Service
A325	Joints, Structural Steel High Strength Bolts For
A475	Wire Strand, Zinc Coated Steel
B2	Wire, Copper, Medium Hard Drawn
B415	Hard-Drawn Aluminum-Clad Steel Wire
B416	Concentric Lay Stranded Aluminum Clad Steel Conductors
D2303-64T	Test For Liquid - Contaminant, Inclined Plane Tracking And Erosion Of Insulating Materials

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70-1981	National Electrical Code

(Application for copies should be addressed to the National Fire Protection Association, 470 Atlantic Avenue, Boston, MA 02210.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Detail specifications. The individual antenna requirements shall be as specified herein and in accordance with the applicable detail specification. In the event of any conflict between the requirements of this specification and the detail specification, the latter shall govern.

3.2 General. The antenna shall be in accordance with MIL-E-16400, to the extent specified herein.

3.2.1 First article. When specified, a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.3 Parts and materials. Parts and materials shall conform to the requirements specified in 3.3.1 through 3.3.8. Parts and materials which are used but which are not specified in 3.3.1 through 3.3.8 shall be in accordance with MIL-E-16400. Parts and materials which are not permitted by MIL-E-16400 shall not be used. Derating and interchangeability shall be in accordance with MIL-E-16400. Equipment performance shall not be dependent on the selection of special values for individual parts. Used or damaged parts or materials shall not be used. All materials shall be fungus inert in accordance with MIL-E-16400.

3.3.1 Towers. Towers shall be as specified in 3.3.1.1 through 3.3.1.2.

3.3.1.1 Structural steel. All structural steel shall conform to the specifications listed under the AISC Specification for the Design, Fabrication, and Erection of Structural Steel for Buildings, unless otherwise specified herein.

3.3.1.1.1 Steel connections. All steel connections shall be made with hardware conforming to 3.5.2.1, 3.5.2.3, or shop welded in accordance with AWS D1.1. Structural connections made with hardware conforming to 3.5.2.3 shall be installed by the turn-of-the-nut method in accordance with Part 5, Specifications and Codes of the Manual of Steel Construction.

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3.3.1.2 Structural aluminum. Structural aluminum shall conform to the specifications listed under the Aluminum Construction Manual, "Specification for Aluminum Structures."

3.3.2 Wooden poles. Wooden poles, when specified in the applicable detail specification (see 3.1) shall conform to MIL-P-28766. Splices in wooden poles are not permitted. All poles shall be supplied in single, continuous lengths. Poles shall be designed to have pole bases completely encased in concrete and embedded in the ground. The concrete encasement shall extend from at least 76.2 millimeters (mm)(3 in.) above ground level to not less than 52.4 mm (6 in.) below the bottom of the pole.

3.3.3 Cables. Cables shall be as specified in 3.3.3.1 through 3.3.3.5.

3.3.3.1 Steel cables. Steel cables shall be stranded and shall conform to the requirements of ASTM A475.

3.3.3.2 Aluminum clad steel wire. Aluminum clad steel wire shall conform to ASTM B415. Aluminum clad steel cable shall conform to ASTM B416.

3.3.3.3 Nonmetallic guys. Nonmetallic materials for use as tower and pole support guys shall be limited to materials having physical, mechanical, and electrical properties specified in 3.3.3.3.1 through 3.3.3.3.8.

3.3.3.3.1 Dimensions. The nominal diameter of the guy shall be as specified by the designer, but in no case shall the nominal diameter be less than 9.53 mm (0.375 in.).

3.3.3.3.2 Breaking strength in tension. The minimum breaking strength in tension of the rods with end fittings attached shall be:

Rod diameter mm (in.)	Minimum breaking strength, kilograms (kg) (pounds (lbs))
9.53 (0.375)	5455 (12,000)
12.70 (0.50)	9091 (20,000)
15.88 (0.625)	13,636 (30,000)
19.05 (0.750)	18,182 (40,000)
25.4 (1.0)	29,545 (65,000)

3.3.3.3.3 Elongation. The maximum elongation at one-third the minimum breaking strength shall be less than 1 percent of the length of the rod.

3.3.3.3.4 Wet strength. The wet strength of the rod shall be not less than 90 percent of the average breaking strength specified in 4.5.3.3.

3.3.3.3.5 Fatigue strength. The fatigue strength of the rod, with end fittings installed, shall be not less that 80 percent of the average breaking strength of that size as specified in 4.5.3.3.

3.3.3.3.6 Fittings and fitting attachments. The ends of the guys shall be terminated with galvanized steel wire rope sockets conforming to Type A and Type B, Finish 2 of RR-S-550 of size appropriate to the guy diameter. Cotter pins used with steel fittings shall be Class 302 or Class 304 corrosion-resistant steel conforming to QQ-S-763. The end fittings shall be affixed to the guys and shall be capable of developing 100 percent of the rated minimum breaking strength of the guy.

3.3.3.3.7 End grips. Field installable end grips, if utilized, shall be capable of developing 100 percent of the rated minimum breaking strength of the guy when tested in accordance with 4.5.3.1.

3.3.3.3.8 Electrical resistance. The guys shall have a minimum resistance of 10 megaohms per foot (ft) of length measured by a 500 volt (V) megger under standard room conditions.

3.3.3.4 Cable tolerances. Tolerances on cable elements (see 6.4.3) of the antenna system, including catenaries and radiating elements, both for the lengths of the cables and for the locations of the fittings and connections along the cable, shall not exceed the values specified herein but may be less if dimensional stability of the antenna system requires it. Tolerances shall be referenced to the cable end points, shall be noncumulative, and shall be measured under a minimum of 10 percent prestress of maximum design load.

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Total length of cable (meters (m) (ft))	Maximum tolerance mm (in.)
30.48 (over 100)	6.35 (0.25)
3.048 to 30.48 (10 to 100)	3.18 (0.125)
3.048 (under 10)	1.59 (0.625)

These tolerances do not apply to the overall length of a cable containing a turnbuckle or other length adjusting device, nor to structural guy insulator locations.

3.3.3.5 Turnbuckles. Turnbuckles terminating at the same anchor point shall be separated so that they turn without interference.

3.3.4 Flat washers. Flat washers shall be used to avoid marring the protective finishes of parts being joined by threaded devices. This requirement applies whether connections are made in the factory or in the field.

3.3.5 Insulating material. Insulating material for radio frequency (RF) applications shall be selected in the descending order of category preference specified in a through d:

a. Category A: Vitrified ceramics in accordance with MIL-I-10 in Grades L4 or better of stearite, alumina, and forsterite only
b. Category B: Other vitrified ceramics complying with MIL-I-10
c. Category C: Arc-resistant materials listed in MIL-STD-454, Requirement 26
d. Category D: Insulating materials in accordance with MIL-STD-454, Requirement 11

All ceramic insulators shall be glazed. Insulators other than ceramic shall be adequately protected against tracking, erosion, ultraviolet radiation, fungus, and water absorption by utilizing a silicone rubber coating or by other suitable means. Insulator material for use on transmitting antennas, (Power 0 through Power 50) (see 1.2.1), selected from other than Category A and Category B (with coating) shall be subject to successful completion of the Inclined Plane Test of ASTM D2303, as modified and specified in 4.5.2.

3.3.5.1 Structural guy cable (see 6.4.1) strain insulators. Structural guy cable strain insulators shall be in accordance with ANSI C29.1 to C29.9 inclusive, Class 54 in the size and strength needed to conform to the requirements of this specification and the applicable detail specification (see 3.1). Where strength requirements exceed that available in this class, open end insulators in conformance with MIL-I-17918 may be used.

3.3.6 Selection of parts. Selection of parts and approval of nonstandard parts shall be in accordance with MIL-E-16400, except that approval of nonstandard parts is not required for structural items, but rather is only required for electrical, electromechanical, and electronic items.

3.3.6.1 Structural items. For nonstandard parts determination purposes, structural items include: structural steel, aluminum and wooden members, structural cables, guys and guy grips, anchors, fastening hardware, antenna or ground screen wire, and similar items of an inert, structural nature.

3.3.6.2 Parts control. Electrical and mechanical parts to be incorporated in the equipment shall be controlled in accordance with MIL-STD-965, Procedure I.

3.3.6.2.1 Electrical and electronic items. For nonstandard parts purposes, these parts include but are not limited to: RF insulators, electronic components such as semiconducters, resistors, capacitors, inductors, transformers, electrical connectors, cables, and similar items; components contained in a matching unit; components contained in the obstruction lighting system; and any similar parts which are basically of an electrical, electromechanical, or electronic (as contrasted to a structural) nature.

3.3.7 Dissimilar metals. Dissimilar metal contacts shall be limited to those specified in MIL-E-16400, under the condition for metal combinations subject to spray from sea water, unless otherwise specified herein. If it is necessary to assemble dissimilar metals other than those allowed by MIL-E-16400 in intimate contact with each other, approval shall be obtained from the procuring activity and the metals shall be protected against electrolytic corrosion in accordance with the methods specified in MIL-E-16400. If electrical considerations preclude the use of such methods, the metal contact shall be protected using the compound specified in 3.5.5. Dissimilar metal connections shall not be required to be made in the field.

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3.3.7.1 Connections between copper and aluminum or galvanized steel. When connections between copper and aluminum or galvanized steel cannot be avoided, the copper shall be tinned and the connection isolated from the environment using the methods specified in 3.5.5.

3.3.8 Capacitors. Capacitors shall be in accordance with the Capacitors paragraph of MIL-E-16400.

3.4 Assemblies. Assemblies shall be as specified in 3.4.1 through 3.4.4.

3.4.1 Metallic cables. Metallic cable assemblies (see 6.4.2) shall be prefabricated from stranded steel cable, stranded aluminum, or stranded aluminum-clad steel cable, segmented by strain insulators where necessary. The length of metal between and beyond insulators shall conform to the performance requirements specified herein. Cable loops shall be secured by field installable end grips, automatic jaw type fittings, or preassembled factory-installed swaged fittings. The fittings shall be capable of developing 100 percent of the minimum breaking strength of the cable. Swaged fittings shall not be required to be made in the field during installation.

3.4.1.1 Metallic cable site accommodations. Metallic cable assemblies shall be designed to accommodate any site condition ranging from flat and level to that specified in 3.9.1.15 and shall be provided in sufficient quantity to accommodate these conditions.

3.4.2 Nonmetallic guys. Nonmetallic guys shall not be assembled in segments but shall be furnished in continuous lengths in accordance with 5.1.1 except that the anchor ends of the guys shall be short lengths of metallic cable conforming to 3.3.3 and 3.4.1, as necessary, to attach tensioning devices that may otherwise damage the nonmetallic guys during and after erection of the antenna. The requirements of 3.4.1.1 and 3.14.1 shall apply to connections of metallic and nonmetallic guys.

3.4.3 Towers. Except as specified in 3.4.3.1 and 3.4.3.2, towers shall not be prefabricated in sections. Rather, tower legs, cross braces, and other tower hardware shall be provided disassembled, for assembly in the field. Field personnel shall not be required to assemble sections in excess of 6.096 m (20 ft) each.

3.4.3.1 Tower fabrication sections. Towers may be fabricated in welded sections not exceeding 6.096 m (20 ft) in length. The use of welded prefabricated towers shall be limited to receiving antennas, (Power 0) (see 1.2.1), when necessary to reduce noise and intermodulation products.

3.4.3.2 Tower welding. Towers may be welded on two sides so that towers can be packaged in a nested fashion suitable for shipping.

3.4.4 Tower legs. Tower legs shall be constructed so that water cannot accumulate inside the structure.

3.5 Processes. Except as specified in 3.5.1 through 3.5.8, processes shall be in accordance with MIL-E-16400.

3.5.1 Structural steel. All structural steel shall be hot dip galvanized after fabrication in accordance with ASTM A123.

3.5.2 Steel hardware. Steel hardware shall be as specified in 3.5.2.1 through 3.5.2.5.

3.5.2.1 Steel bolts, nuts, and washers. Steel bolts, nuts, and washers, when in contact with galvanized steel, shall be either galvanized in accordance with ASTM A153 or coated by the SERMETEL process, or equal, except that high strength steel shall not use the SERMETEL process.

3.5.2.2 Steel bolts, nuts, and washers, other than high strength steel. Steel bolts, nuts, and washers other than those made of high strength ASTM A325 steel, when used in contact with aluminum, shall be hot dip aluminum coated in accordance with MIL-C-83488, Class I, or AWS C2.2, or made of Type 316 corrosion-resistant steel.

3.5.2.3. Steel bolts, nuts, and washers, high strength steel. Bolts made of high strength ASTM A325 steel other than those in contact with aluminum shall be galvanized in accordance with ASTM A153, Class C. Steel nuts and washers used with galvanized ASTM A325 steel bolts shall be made of ASTM A194, Grade 2H steel and galvanized in accordance with ASTM A153, Class C. Bolts, nuts, and washers of ASTM A325 steel in contact with aluminum shall be aluminum coated in accordance with AWS C2.2.

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3.5.2.4 Steel hardware other than bolts, nuts, and washers. Steel hardware other than bolts, nuts, and washers, when used in contact with aluminum surfaces shall be either galvanized in accordance with ASTM A153 or aluminum coated. Where an assembly of galvanized steel hardware other than bolts, nuts, and washers is used in contact with an aluminum surface, an interposing coating shall be applied to the contact surface of the galvanized steel. This coating shall consist of a primer coat conforming to TT-P-641 (zinc dust - zinc oxide, Type I). All threaded members shall have their junctions and thread projections coated with an environmental sealing compound in accordance with MIL-S-38228, Type I-2, after the members have been assembled. The primer may be omitted from the threaded portions if primer interferes with threading action.

3.5.2.5 Cadmium plating. Cadmium plating shall not be used.

3.5.3 Nonmetallic cables. Nonmetallic cables shall be provided with protection against deterioration from exposure to ultraviolet radiation and weathering and reduction of electrical resistance due to moisture absorption

3.5.4 Stranded steel cable elements. Stranded steel cable elements shall be zinc coated in accordance with ASTM A475, Class C.

3.5.5 Protective compounds for dissimilar metal contacts. Compounds for protecting dissimilar metal contacts (see 3.3.7) shall conform to MIL-C-16173. Compounds for treating electrical connections (see 3.7.1) shall conform to MIL-C-16173 or be DOW Corning Silastic 738 RTV, or equal.

3.5.6 Castings. Castings shall be in accordance with Requirement 21 of MIL-STD-454. Die casting may be used only for aluminum enclosures.

3.5.7 Welding and soldering. Welding of structural members shall be in accordance with the applicable requirements of the American Welding Society specifications. Welding and soldering of electrical connections shall be in accordance with MIL-E-16400, except that approval of the welding techniques used to make electrical connections is not required. Silver solder shall comply with QQ-B-654, Grade III. Corrosive flux shall not be used.

3.5.8 Brazing. Brazing shall conform to Requirement 59 of MIL-STD-454.

3.6 Design objectives. Design objectives shall be in accordance with 3.6.1 through 3.6.3.

3.6.1 Service life. The antenna shall be designed and constructed to provide continuous service for long periods of time. Mechanical and electrical integrity shall be retained for not less than 15 years without replacement of major components or assemblies. Routine inspection and minor adjustments shall not be required at intervals of less than 12 months and shall not require a downtime of more than 3 hours. Routine inspection shall not require lowering the antenna. The design, choice of materials, parts selection, construction methods, surface protection, and other factors which determine durability and performance reliability shall be consistent with this requirement when interpreted under the most severe combination of environmental conditions specified in 3.13.

3.6.2 Moisture pockets. The antenna shall conform to Requirement 31 of MIL-STD-454.

3.6.3 Erection. The antenna shall be designed so that it can be installed in 7 days by a crew of six men working 8 hours per day (excluding grading, foundation, and anchor installation).

3.7 Electrical design. Electrical design for the antenna shall be in accordance with the requirements specified in 3.7.1 through 3.7.14 and the applicable detail specification (see 3.1).

3.7.1 Electrical bonding. Electrical bonding shall be in accordance with MIL-HDBK-419 and as hereinafter specified. If the protective finishes are omitted at the bond, the bond shall be treated with a compound or sealant conforming to 3.5.5 after assembly.

3.7.2 Corona and electrical breakdown prevention. Corona and electrical breakdown prevention shall be employed in accordance with Requirement 45 of MIL-STD-454.

3.7.3 Obstruction lighting. The electrical design requirement for obstruction lighting and associated control circuitry shall be as specified in 3.15.2.1. Chassis within the enclosure shall be bonded to the enclosure.

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3.7.4 Ground system. All monopole antennas and monopole antenna arrays shall be supplied with an artificial ground system. Dipole antennas and antenna arrays shall not depend upon an artificial ground system to obtain specified performance, unless specifically required by the applicable detail specification (see 3.1). Specific ground system requirements shall be as specified in the applicable detail specification (see 3.1).

3.7.4.1 Material. All ground systems shall consist of No. 10 American Wire Gage (AWG) size as a minimum, for mechanical considerations. The wire shall be medium drawn, solid copper in accordance with ASTM B2. All ground rods shall be Class 1 of W-R-550. All junctions (wire-to-wire, wire-to-ground rod, or wire-to-antenna) and all wire crossovers shall be bonded in accordance with 3.7.12. The ground system need not be preassembled.

3.7.5 RF insulators. The type, configuration, and grade of insulator shall be as dictated by the particular application, the environmental conditions, and the requirements of 3.5.5.

3.7.5.1 Insulator application. The application of insulators shall be such that moisture accumulation does not occur at or on the insulator. Insulators, except those used on structural guys, shall be designed to minimize the possibility of natural or gravity fed moisture paths caused by condensation, precipitation, or melting ice on the antenna.

3.7.6 Input impedance. The antenna input impedance at the RF connector shall be 50-ohms, unbalanced. Unless otherwise specified in applicable detail specification (see 3.1), the voltage standing wave ratio (VSWR) shall not exceed 2 to 1 with respect to 50 ohms. If an impedance matching transformer balanced or unbalanced (balun) is necessary to comply with this requirement, the transformer shall be in accordance with MIL-T-28732, except for those portions specifically waived in the contract (see 6.2). The input impedance and VSWR shall then apply to the input of the transformer when it is connected to the antenna. If a matching unit or network other than a balun transformer is used, the unit or network shall conform to the applicable requirements of MIL-T-28732 and the requirements of this specification.

3.7.6.1 Factors of safety. The specified min factors of safety shall be applied to the design of the components indicated:

For	As to	Safety factor
Transformer	Flux density	1.4
	Voltage	3.5
	Current	1.5
Inductors	Current	1.5
	Voltage	3.5
Capacitors	Current	1.5
	Voltage	4.0

3.7.6.2 Nominal secondary impedance. If a balun transformer is used in the design, the impedence interface between the balanced terminals on the balun transformer and the antenna feed line shall be one of the following impedances only: 200 ohms, 300 ohms, or 600 ohms. Balun transformer impedances shall not be complementary to the antenna feed line impedance such that a special selection procedure for replacement balun transformers is required. The feedline and balun transformers impedance characteristics shall be such that the balun transformer can be replaced with any balun transformer conforming to MIL-T-28732, as applicable, having the same impedance regardless of source.

3.7.6.3 Lumped constants. Discrete lumped constants, such as capacitors and inductors, shall not be used to obtain artificial foreshortening or phase shifting of individual radiation elements without specific written approval of the procuring activity. This requirement does not preclude the use of such components in the input matching unit.

3.7.7 RF input connectors. All connectors shall be female receptacles and shall contain a gas barrier capable of withstanding 30 pounds per square with gage (psig) and shall be located at the base of the antenna. If an impedance matching transformer is used, input connectors shall be as specified in MIL-T-28732. If a matching transformer is not used, the input connector shall be as specified in TABLE I.

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TABLE I. Connector power and size.

Power (see 1.2.1)	Connector type, size	Applicable specification
0	N	MIL-C-39012
1	LC	MIL-C-3650
5, 10	41.28 mm (1.625 in.)	MIL-T-28732
10a, 25	79.38 mm (3.125 in.)	MIL-T-28732
50	155.58 mm (6.375 in.)	MIL-T-28732

3.7.8 Lightning protection. Unless precluded by RF design considerations, two legs of each antenna support tower used in the design shall be grounded through two separate ground rods. This grounding shall be separate and independent from the RF ground system specified in MIL-HDBK-419. Antennas not utilizing a matching unit which already has its secondary protected against lightning shall have nonconsummable, self-extinguishing gaps protecting each conductor of the RF input. One side of the gaps shall be grounded. The gaps shall be capable of conforming to the requirements of MIL-A-9094 for stroke currents and repeated operation, and shall not degrade the antenna below the requirements of this specification and the applicable detail specification (see 3.1).

3.7.8.1 Ground rod. Ground rods, lightning rods, and connectors required for grounding shall be provided as specified in a through c:

a. The rods shall be Type I, Class B of W-R-550, with the exception of 15.87 mm (0.625 in.) diameter and 1.8288 m (6 ft.) length

b. Connections from the lightning rod and to the ground rod shall be by size AWG No. 0000, or larger copper wire, utilizing copper or copper clad clamps where necessary. Braided straps shall not be used.

c. Connection wires between aluminum and galvanized towers to copper and copper clad ground rods shall be tinned at the tower end over one-half of its length.

3.7.9 Direct current (DC) continuity to ground. The antenna shall have a DC continuity to ground of less than 5 ohms. This continuity may be accomplished inherently through an impedance matching transformer or by a static drain coil. The system providing DC path to ground shall be protected by the ball gap specified in 3.7.8, and shall be capable of withstanding twice the ball gap breakdown voltage. The provision of DC continuity and the protecting ball gap shall not degrade the performance of the antenna below the requirements of this specification and the applicable detail specification (see 3.1).

3.7.10 Power input capability. The antenna shall be capable of handling the average and peak instantaneous RF input power without long term degradation, and without arcing, corona, or detrimental heating when the power specified in 1.2.1 is applied to the RF connector. The rating is for continuous duty under all possible combinations of VSWR and performance parameters specified herein and in the applicable detail specification (see 3.1).

3.7.11 Additional performance requirements. Additional performance requirements, such as, but not limited to, a through e, shall be as specified in the applicable detail specification (see 3.1):

a. Gain (see 6.4.4)
b. Efficiency (see 6.4.5)
c. Polarization
d. Frequency range
e. Radiation patterns

3.7.12 Electrical connections at RF. The antenna shall be designed so that the number of electrical connections at RF to be made in the field is a minimum. All electrical metal-to-metal connections which are part of the RF circuit, including the radiating elements, feed line, and ground system shall be as specified in 3.7.12.1 through 3.7.12.6.

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3.7.12.1 Connection during manufacture. Connections made during manufacture shall be by compression (swage) fittings, bolts, welding, or brazing at the factory.

3.7.12.2 Connections in the field. Connections to be made in the field shall be as specified in a and b:

a. Electrical connections shall be bolted with the exception of the RF ground system. Connections shall conform to 3.7.1.

b. All ground system connections shall be made by brazing conforming to MIL-STD-454, Requirement 59, including connection of the ground system to the base of the antenna and to the grounding rods. Every point where a ground wire must be attached shall have a means for mechanically anchoring the wire prior to welding or soldering so that the weld or solder joint is not depended upon for mechanical strength.

All parts, materials, tools (other than common hand tools) (see 6.6), and sealants necessary to make field connections as specified in a and b shall be supplied with the antenna and in sufficient quantity to allow at least 110 percent of the actual number of connections to be made. Under no circumstances shall swage type fittings be required to be compressed in the field. All clamped connections shall be secured by at least two bolts.

3.7.12.3 Turnbuckles and clevises. Turnbuckles, clevises, and other nonrigid or adjustable connectors, which would be detrimental to the continuity of or generate noise in the RF electrical circuit, shall be isolated from the circuit with insulators, and jumpered when continuity is required.

3.7.12.4 Electrical connection degradation protection. Electrical connections at RF shall be constructed and protected so that the antenna system will not gradually degrade below the requirements specified herein and in the applicable detail specification (see 3.1) over its service life due to deterioration of the connection (for example, increase in contact resistance, heating, high VSWR, corrosion, and so forth). Aluminum-to-aluminum and copper-to-copper connections which are not welded, brazed, or swaged shall be made as specified in a through d:

a. The surfaces to be mated shall be coated with Alco No. 2 Electrical Joint Compound, or antiseize compound conforming to MIL-T-22361 after both surfaces have been cleaned.

b. The surfaces shall be abraided through the compound with a wire brush or abrasive cloth.

c. The surfaces shall be joined and the excess compound shall be wiped off.

d. The joined surfaces shall be sealed completely from the environment in accordance with 3.5.5.

This procedure applies to both factory-made and field-made connections.

3.7.12.5 Electrical continuity. All electrical connections shall be thoroughly cleaned of oxide formation, grease, and other contaminants which would be detrimental to electrical continuity prior to assembly. Threads of fastening devices shall not be depended upon for electrical continuity.

3.7.12.6 Bolted RF connections. Bolted RF connections to be made between metallic surfaces in the field shall require the use of conductive antiseize compound conforming to MIL-T-22361.

3.7.13 Additional noise reduction considerations. When a receive only antenna is specified (Power 0) (see 1.2.1), special attention in design shall be directed toward reduction of antenna noise. As a minimum, the techniques specified in 3.7.13.1 shall be employed.

3.7.13.1 Metal-to-metal connections. Metal-to-metal connections which are not part of the RF circuit, but due to their proximity generate antenna noise, shall be welded, or securely bolted, or clamped. Connections which must retain flexibility or adjustability due to mechanical considerations (such as turnbuckles and clevises) shall be jumpered.

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3.7.14 Intermodulation and harmonic products. Second and third order harmonic and intermodulation products, when one or two equal power input signals are applied to the RF input connector so that the antenna is driven at the power level specified in 1.2.1, shall not exceed 60 dB below the combined power level. Second and third order harmonic products of frequency f_1 are $2f_1$ and $3f_1$; second and third order intermodulation products of frequencies f_1 and f_2 are $f_1 \pm f_2$, $2f_2 \pm f_1$, and $2f_1 \pm f_2$. Frequencies f_1 and f_2 shall be within the frequency range of the antenna, but their products shall not be so constrained.

3.8 Mechanical design and construction. Mechanical design and construction shall be as specified in 3.8.1 through 3.8.3.

3.8.1 Enclosures. All enclosures shall be totally encased as specified in MIL-STD-108.

3.8.2 Lifting attachments. Lifting attachments shall be provided as required to provide a point of attachment for every erection cable to be used during the antenna installation or in a normal maintenance procedure. Lifting attachments shall be permanently attached to the structure and shall be designed with a safety factor of a least 4.

3.8.3 Corners and edges. Corners and edges shall be rounded in accordance with MIL-E-16400.

3.9 Structural requirements. Structural requirements shall be as specified in 3.9.1 through 3.9.1.15.

3.9.1 General design criteria. The antenna shall be designed in accordance with the criteria specified in 3.9.1.1 through 3.9.1.15. Support poles and guyed towers shall be designed as beam columns on elastic supports.

3.9.1.1 Moment of inertia. Moment of inertia may be assumed to be constant within a span.

3.9.1.2 Axial load. Axial load may be assumed to be constant within a span.

3.9.1.3 Uniform variation of wind pressure. Uniform variation of wind pressure may be replaced by a step function that makes the wind load constant in a span.

3.9.1.4 Wind velocity. Wind velocity to be used for computing wind loads on guys may be assumed as the wind acting at mid-height of the guys.

3.9.1.5 Wind loading on the supporting poles or towers. The wind pressures on the supporting poles or towers shall be computed in accordance with the methods specified in NAVFAC DM-2.2 for wind pressure.

3.9.1.6 Wind loading on cable elements of the antenna system. Wind, drag, and lift coefficients on all cable and antenna elements of the system shall be computed in accordance with the methods specified in NAVFAC DM-2.2 for wind forces on guy wires and cables.

3.9.1.7 Combinations of loads. The antenna shall be designed for the combinations of loads with corresponding allowable stresses specified in TABLE II.

TABLE II. Combinations of loads.

Load combinations	Percentage of allowable stress
Dead load plus wind load	100
Dead load plus wind load and ice load	100
Dead load plus one-quarter wind load plus the effect of failure of any one guy at any guy level	133 1/

1/ But not to exceed yield strength of the material

3.9.1.8 Ice. Whenever ice is considered in the design, the consideration shall be applied to all cable and antenna elements and to all the supporting structures of the antenna system.

3.9.1.9 Maximum cable stresses (metallic cable). Maximum cable stresses in metallic cable elements of the antenna system shall be not more than 50 percent of the minimum breaking strength of the cable.

3.9.1.10 Maximum cable stresses (nonmetallic cable). Maximum stresses in nonmetallic guy elements of the antenna system shall be not more than 30 percent of the minimum breaking strength or as specified in 4.5.3.3, whichever is lower.

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3.9.1.11 Insulators. Maximum design stress in an insulator shall be not more than 30 percent of the guaranteed static breaking strength of the insulator, with the exception that the eye bolt of a yoke type insulator shall be designed for stresses not more tha one-fifth of the minimum breaking strength.

3.9.1.12 Structural steel allowable stresses. Structural steel allowable stresses shall be in accordance with the applicable sections of the AISC specifications for the Design, Fabrication, and Erection of Structural Steel for Buildings.

3.9.1.13 Structural aluminum allowable stresses. Structural aluminum allowable stresses shall be in accordance with specifications for Aluminum Structures, Aluminum Construction Manual.

3.9.1.14 Allowable stresses for wooden poles. Allowable stresses shall be in accordance with the National Forest Products Association National Design Specifications for stress-grade Lumber and its fastenings, except that the section for allowable unit stress for short time loading shall not apply.

3.9.1.15 Foundation and anchor design. Foundation and anchor design shall be in accordance with NAVFAC DM-7.2. Aluminum shall not be required to be embedded in the earth or in concrete. Hardware to be embedded in concrete (such as anchor bolts, pins, and so forth) shall be held to tolerance of less than 12.70 mm (±0.50 in.). The contractor shall supply templates to ensure placement of the hardware to the accuracy required. Unless otherwise specified in the detail specification (see 3.1) or in the contract (see 6.2), site grading shall not be required to less than 609.6 mm (±24 in.) flatness and less than ±3 degrees in levelness in order to obtain specified performance with all poles and towers remaining plumb (vertical) and arbitrary orientation of the antenna (see 3.1).

3.10 Maintenance design. Maintenance design shall be as specified in 3.10.1 and 3.10.2.

3.10.1 General purpose electronic test equipment (GPETE). For purposes of maintenance and necessary repair, GPETE shall be selected from MIL-STD-1364.

3.10.2 Insulator replacement. All insulators used in the antenna shall be field replaceable within 15 minutes after access (not including time to reach to the insulator). Insulators within 0.9144 m (3 ft) of support members such as towers and poles, and those within 4.572 m (15 ft) of the ground level shall be replaceable without lowering the antenna or slackening (antenna) tension if the insulators connect to radiating elements.

3.11 Equipment to be furnished by the contractor. Equipment furnished by the contractor shall include, but is not limited to, all towers, poles, guys and anchors, hardware, fittings, antenna components, repair kits, and special tools (see 6.6) necessary to provide a complete operational antenna. Concrete, reinforcing rods, and earth type material such as gravel or sand are not required. An RF electrical ground system shall be provided when required to obtain specified performance. In so far as practicable, the maximum amount of preassembly and prefabrication shall be provided, with necessary adjustments to ensure dimensional control and ease of field installation.

3.11.1 Start up spares. In addition to the hardware furnished for assembly and erection of the antenna, parts such as bolts, nuts, washers, thimbles, clevices, pins, wire lugs, crimp connectors and insulators, tower cross braces, and similar items subject to loss, wear, or breakage shall be supplied in excess quantity of 10 percent of each type. A minimum of one additional of each size and type shall be supplied. If wire lugs and crimp connectors are part of the antenna design, a crimping tool or tools shall be supplied to allow on-site repairs.

3.11.2 Field change kits. If field changes are necessary to correct equipment defects during the period of the contract or the warranty, the changes shall conform to MIL-F-17655, Type I, Class C.

3.11.3 Special tools. The requirement for special tools shall be as specified in MIL-STD-454, Requirement 63 (see 6.6).

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3.11.4 Installation kit. An installation kit shall be supplied with each antenna system and shall contain the items specified in a through f:

a. A 227-grams (g) (8-ounce (oz)) supply (minimum) of each coating and treatment used in the antenna as specified in 3.5.5, 3.7.12.4, and 3.7.12.6, in addition to that required for installation, and to provide for loss, spillage, and maintenance

b. A wire brush or abrasive cloth as required to comply with the requirements of 3.7.12.4 in the field

c. A touch-up kit containing 445g (16 oz) each of primer and paint, and an applicator for touching up all coatings other than those in a, which are specified herein, and which are used on the antenna. Galvanizing repair kits shall comply with MIL-P-21035.

d. Special tools required by 3.11.3

e. Start up spares required by 3.11.1

f. Templates for location of tower hardware embedded in concrete

3.12 Thermal design. Thermal design shall be in accordance with the Thermal design paragraph of MIL-E-16400.

3.12.1 Dielectric heating. All components, including wooden poles, shall be capable of withstanding voltage gradients produced by the antenna in operation at maximum power without damage.

3.13 Environmental conditions. The antenna shall provide underground operational performance when exposed to the environmental conditions specified in 3.13.1 and 3.13.5. This requirement is for operation and not merely survival (see 1.2.1).

3.13.1 Humidity. The humidity requirements of MIL-E-16400 shall apply. Wind and icing shall be as specified (see 1.2.2).

3.13.2 Salt fog spray. The antenna shall be capable of normal operation under the salt fog environment encountered in the coastal regions of the world. The antenna shall not be degraded either structurally or electrically after a salt fog test is performed in accordance with MIL-E-16400 and 4.5.10.3. The DC contact resistances of electrical connections shall not increase by more than 15 percent after completion of the salt fog test.

3.13.3 Thimbles. Thimbles shall be used at all guy connections as needed.

3.13.4 Electromagnetic pulse (EMP). The system shall withstand the EMP resulting from an exoatmospheric nuclear explosion, whose RF field is:

$$E(t) = E_0(e^{-\alpha t} - e^{-\beta t}) \text{ and}$$

$$H(t) = \frac{E(t)}{Z_0}, \text{ where}$$

$E_0 = 5.25 \times 10^4$ volts/meter

$\alpha = 4.0 \times 10^4$ per second

$\beta = 4.8 \times 10^8$ per second

$Z_0 = 377$ ohms

3.13.5 Vibration. The antenna shall be designed to eliminate natural resonances which can be excited by the specified environmental conditions and which will be detrimental to the operation, structural integrity, and service life of the antenna.

3.14 Identification and marking. Identification and marking shall be in accordance with MIL-E-16400.

3.14.1 Identification of cable connections. Cable segments connected to each other with the same connector shall be so marked that visual inspection of the connection can be made to verify that the corresponding cable segments are connected to each other. This requirement applies to both factory-installed and field-installed connections.

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3.15 Safety. Safety shall be in accordance with the Safety criteria paragraph of MIL-E-16400 and as specified in 3.15.1 through 3.15.6.2.

3.15.1 Climbing equipment. Towers over 6.096 m (20 ft) in height shall be provided with climbing equipment consisting of a ladder conforming to 3.15.1.2, and a climbing safety device, conforming to 3.15.1.1, for the full height of the tower.

3.15.1.1 Safety climbing device. The safety climbing device shall consist of the features specified in a through d:

a. A rigid carrier rail
b. A means of attaching the rigid carrier rail to the structure
c. A locking mechanism or safety sleeve which travels on the carrier rail
d. A belt to connect the workman to the locking mechanism

The complete antenna shall be in conformance with the requirements of RR-S-1301, Type I. The antenna shall allow the worker to operate freely in a normal climbing position during ascent or descent and shall enable the worker to be attached to the device at all times during the climb or descent without having to remove his hands from the structure to operate the system effectively.

3.15.1.2 Ladders. Ladders shall be constructed of the same material as the tower and shall conform to the requirements specified in 3.3.1.1 through 3.3.1.2. Ladders shall consist of either factory-assembled sections of lengths consistent with the tower section sizes, or shall be integrated in the tower using the tower legs as stringers and the tower's horizontal cross bracing members as rungs.

3.15.1.2.1 Dimensions. The inside dimensions between vertical stringers shall be uniform and shall be not less than 406.4 mm (16 in.) (see FIGURE 1). Center-to-center placing of the rungs shall be uniform and shall be not less than 254.0 mm (10 in.) nor more than 304.8 mm (12 in.) (see FIGURE 1). Vertical stringers for separate ladders shall be 63.5 mm (2.5 in.) wide by 9.525 mm (0.375 in.) thick minimum. Angle brackets for bolting separate ladders to the tower shall have holes for bolts for connection to the bracing members of the tower and shall be 9.525 mm (0.375 in.) thick, minimum and spaced not more that 3.048 m (10 ft) apart.

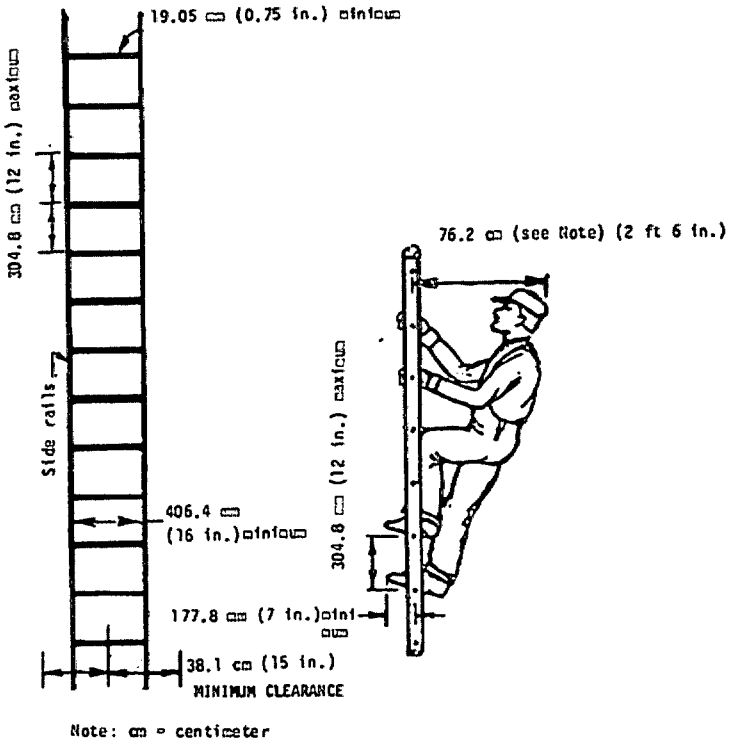


FIGURE 1. Ladder clearances.

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3.15.1.2.2 Rungs. The rungs shall have a minimum projected step width of 19.05 mm (0.75 in.) (see FIGURE 1).

3.15.1.2.3 Steel ladders. Steel ladders shall be galvanized after fabrication. Splice plates for bolted connections shall be provided at all ends for connecting individual sections.

3.15.1.2.4 Minimum design level load. The minimum design level load shall be a single concentrated load of 909 kg (200 lbs). The number and positions of additional concentrated live load units of 200 lbs each, as determined from anticipated usage of the ladder, shall be considered in the design. The live load imposed by persons occupying the ladder shall be considered to be concentrated at points which will cause the maximum stress in the structural member being considered. The weight of the ladder and attached appurtenances, together with the live load, shall be considered in the design of rails and fastenings.

3.15.1.2.5 Clearances. Clearances shall be as specified in 3.15.1.2.5.1 and 3.15.1.2.5.2.

3.15.1.2.5.1 Climbing side. The perpendicular distance from the center line of the rungs to the nearest permanent object on the climbing side of the ladder shall be 76.20 cm (30 in.) (see FIGURE 1). A clear width of at least 38.10 cm (15 in.) shall be provided each way from the center line of the ladder in the climbing space (see FIGURE 1).

3.15.1.2.5.2 Clearance behind ladder. The distance from the center line of rungs or steps to the nearest permanent object behind the ladder shall be not less than 177.8 mm (7 in.) (see FIGURE 1) except that when unavoidable obstructions are encountered, minimum clearance as shown on FIGURE 2 shall be provided.

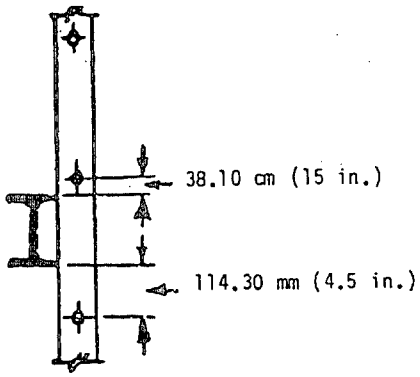


FIGURE 2. Minimum ladder clearances for unavoidable obstructions.

3.15.2 Aircraft obstruction lighting and marking. All antennas higher than 30.48 m (100 ft) above ground level shall have obstruction marking and lighting. The contractor shall furnish the marking and lighting kit form. The kit shall contain all parts and materials required to mark and light the antenna in accordance with FAA Advisory Circular 70/7460-1. The parts and materials shall comply with the Advisory Circular 70/7460-1.

3.15.2.1 Electrical design. The antenna shall conform to the limits of MIL-STD-461 and MIL-STD-462 specified in a and b:

a. Power 0 and Power 1 antennas: RE03
b. Power 1 through Power 50 antennas: RS03

All interconnecting cabling shall be supplied so that only a power source need be at the base of the antenna. The power source shall be as specified for the fixed shore station alternating power source in MIL-E-16400. Internal wiring and cabling, and mounting of parts shall be in accordance with MIL-E-16400. The lighting system shall not cause interference or modulation and shall be designed so that its installation shall conform to the National Electrical Code (see NFPA 70-1981).

3.15.2.2 Control device. A light sensitive control device adjustable in azimuth so that the obstruction lights will be turned on at a north sky light intensity level of not less than 35 foot-candles and turned off at a light intensity level not greater than 58 foot candles shall be provided as part of the lighting system. The control device shall be located at the base of the antenna, no higher than 3.048 m (10 ft) from the ground level.

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3.15.2.3 Power source. The power source shall be 115 8, single-phase 50 or 60 hertz (Hz) and the obstruction lighting shall be designed for that service.

3.15.2.4 Shock. The lighting system shall be capable of withstanding normal bench handling in service.

3.15.3 Locking of turnbuckles. All turnbuckles shall be provided with locking nuts or other positive locking devices.

3.15.4 Guy failure. The antenna shall be designed to withstand the effect of the failure of any guy. The antenna loading and allowable stress criteria for this condition shall be one-quarter maximum wind load plus all dead load with an increase in allowable stress as specified in 3.9.1.7, but not exceeding the yield strength of the material.

3.15.5 Unacceptable materials. Equipment design shall not include the materials specified in a through d:

a. Polychlorinated biphenyls
b. Asbestos and asbestos compounds
c. Flammable or combustible materials
d. Lithium and lithium compounds unless specifically approved by the procuring activity

The Unacceptable materials paragraph of MIL-E-16400 shall apply.

3.15.6 Labeling. Labeling shall be as specified in 3.15.6.1 and 3.15.6.2.

3.15.6.1 Radiation and shock hazards. Radiation and shock hazards shall be labeled. Labels shall be in a position readily visible to personnel.

3.15.6.2 Equipment weight. Equipment capable of being lifted and weighing more than 15.9 kg (35 lbs) shall be labeled accordingly.

3.15.7 Leakage current. Leakage current shall be in accordance with MIL-E-16400.

3.16 Workmanship. Workmanship shall be in accordance with Requirement 9 of MIL-STD-454.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification at any time. Verification will consist of, but is not limited to, a) surveillance of the operations to determine that practices, methods, and procedures of the written quality program are being properly applied, b) Government product inspection to measure quality of the product to be offered for acceptance, c) Government inspection of delivered products to assure compliance with all inspection requirements of this specification. Failure of the contractor to promptly correct deficiencies discovered by him or of which he is notified shall be cause for suspension of acceptance until corrective action has been taken or until conformance of the product to prescribed criteria has been demonstrated.

4.1.2 Quality assurance terms and definitions. Quality assurance terms used in this specification shall be as defined MIL-STD-109.

4.2 Classification of inspections. The inspections specified herein are classified as specified in a through c:

a. First article inspection (see 4.3)
b. Quality conformance inspection (see 4.4)
   1. Production inspection (Group A) (see 4.4.1)
   2. Production control inspection (Group B) (see 4.4.2)
   3. Environmental inspection (Group C) (see 4.4.3)
c. Inspection of preparation for delivery (see 4.7)

4.3 First article inspection. Unless otherwise specified (see 6.2), one unit shall be required for first article inspection. First article inspection shall consist of all examination and testing necessary to determine compliance with the requirements of this specification and the applicable detail specification (see 3.1). First article inspection shall include the inspections specified in TABLE III and in the detail specification.

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TABLE III. Examinations and tests.

Examination or test	Requirement paragraph	Test paragraph	First article inspection	Quality conformance inspection		
				Group A	Group B	Group C
Surface examination						
Parts and materials	3.3, 3.15.5, 3.9	4.5.1, 4.5.3	X	X	X	
Finish	3.5	4.5.1	X	X	X	
Marking	3.14	4.5.1	X	X	X	
Safety	3.15	4.5.11	X	X	X	
Workmanship	3.16	4.7	X	X	X	
Preperformance	3.7.3		X		X	
Performance	3.6	4.5.2.2	X	X		
Humidity	3.13.1	4.5.10.2	X			X
Salt fog	3.13.2	4.5.10.3	X			X
Fungus	3.3	4.5.10.1	X			X
Shock	3.15.2.4	4.5.10.4	X			X
Inclined plane	3.3.5	4.5.2	X			
Proof loading	3.3.5.1, 3.4.1, 3.7.5, 3.8.2, 3.9.1.9 to 3.9.1.11, 3.15.1.1, 3.15.1.2.4, 3.15.4	4.5.4	X			
Installability	3.4, 3.4.3.1, 3.6.3, 3.7.4, 3.8.2, 3.11, 3.11.1, 3.11.4, 3.15.2, 3.15.3	4.5.6	X			
Enclosure	3.8.1	4.5.5	X			
Weld	3.5.7	4.5.7	X	X	X	
Steady-state voltage and frequency	3.15.2.1	4.5	X		X	
Transient voltage	3.15.2.1	4.5	X			
Transient frequency	3.15.2.1	4.5	X			
Spike voltage	3.15.2.1	4.5	X			
Power interruption	3.15.2.2	4.5	X			
Power and power factor	3.15.2.3	4.5	X			
Insulation resistance	3.7.5	4.5	X			
Corona measurement	3.7.2	4.5	X			
Leakage current	3.1	4.5.13	X			
Dielectric withstanding voltage	3.15.2.1	4.5	X			
Electromagnetic interference	3.15.2.1	4.5.12	X			
Thermal design	3.12	4.5.9	X			
Maintainability demonstration	3.10	4.6	X			
Paint system	3.5	4.5.1	X	X		

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4.4 Quality conformance inspection. Quality conformance inspection shall be as specified in 4.4.1 through 4.4.4 and in TABLE III.

4.4.1 Production inspection (Group A). Production inspection shall be conducted on each antenna offered for delivery. The inspection shall comprise such examinations and tests which will prove the workmanship and reveal omissions and errors of the production process, such as functional and performance tests at a limited number of points in the required range, tests which detect deviations from design, tests of adjustments, and tests which detect hidden defects of materials. Production inspection shall include the examinations and tests shown in Group A of TABLE III and in the detail specification (see 3.1).

4.4.2 Production control inspection (Group B). Production control inspection,including sampling, shall conform to the examinations and tests shown in Group B of TABLE III and to the inspection procedures of MIL-STD-105 using the special inspection levels. The inspection level shall be S-3 for normal, tightened, and reduced inspection. Production control inspection shall be performed on antennas which have passed production inspection. Production control inspection shall be approved. The antennas shall satisfactorily conform to the requirements of production control inspection prior to release for shipment.

4.4.2.1 Rejected lots. If an inspection lot is rejected, the contractor may withdraw the lot from further inspection. The contractor may also rework a rejected lot to correct the defective antennas and reinspect the lot using tightened inspection. Rejected lots shall be kept separate from new lots and shall not lose their identity.

4.4.3 Environmental inspection (Group C). Environmental inspection shall be accomplished on a sampling basis as specified in 4.4.3.1. Environmental inspection shall encompass environmental tests to prove the durability of the materials, parts, and the antenna as a whole; tests of the effects of changes of environment (such as extremes of temperature and humidity); and tests of the effects of shock and vibration. Environmental inspection shall include the examinations and tests shown in Group C of TABLE III.

4.4.3.1 Sampling for environmental inspection. One antenna shall be selected from each successive 10 antennas, or fraction thereof, produced. The first sample antenna shall be selected from the first month's production.

4.4.3.2 Nonconforming environmental sample units. If a sample unit fails the inspection specified in 4.4.3, the contractor shall immediately investigate the cause of failure and shall report to the quality assurance representative (QAR) the results thereof and details of the corrective action taken to correct units of product which were manufactured under the same conditions, with the same materials, processes, and so forth. If the QAR does not consider that the corrective action will enable the product to conform to specified requirements, or if the contractor cannot determine the cause of failure, the matter shall be referred to the contracting officer.

4.4.4 Reinspection of conforming production control and environmental test antennas. Unless otherwise specified (see 6.2), sample units which have been subjected to, and have passed, production control inspection may be accepted on the contract provided they are resubjected to, and pass, production inspection after repair of all visible damage.

4.5 Test methods. The examinations and tests specified in TABLE III shall be conducted in accordance with the applicable test methods of MIL-E-16400, except as specified in 4.5.1 through 4.5.13.

4.5.1 Surface examination. Dimensional measurements, quantities, finishes, and workmanship checks shall be performed.

4.5.1.1 Cable measurement. Each cable shall be measured in a prestressed condition to determine compliance with 3.3.3.4 and visually checked for proper assembly and compliance with 3.14.1.

4.5.2 Inclined plane tests. When required by 3.3.5 (that is, use of Category C or Category D material in transmitting applications) test samples of each material shall successfully complete the initial tracking voltage test of ASTM 02303-64T. If the insulating material is to be coated in final use, the test shall be performed on the coated material.

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4.5.2.1 Power supply frequency. The frequency of the power supply used in the test shall be 30 MHz.

4.5.2.2 Limits. The test sample shall not be considered as having passed the test unless:

a. The initial tracking voltage is at least double that which would be encountered under actual use conditions, that is, full specified power operation of the antenna
b. Melting of the material or its coating does not occur

4.5.3 Special tests of nonmetallic guys and assemblies. Certification of previous testing by a contractor will be acceptable in lieu of retesting for the current contract for any of the tests specified in 4.5.3.1 through 4.5.3.4.

4.5.3.1 Breaking strength in tension. The actual breaking strength shall be determined on samples at least 1.2192 m (4 ft.) in length including fittings. The sample shall be pulled in tension at the rate of 50.88 mm (0.2 in.) per minute. The test shall be performed in a room maintained at 70° Fahrenheit (F) to 75° F and relative humidity no higher than 64 percent. A minimum of five samples of each size with fittings attached shall be tested and no sample shall have a breaking strength lower than the rated minimum strength of the sample.

4.5.3.2 Wet stength. Wet strength shall be determined by immersing the sample in boiling water for 3 hours ±10 minutes. The sample shall then be allowed to return to room temperature by immediately immersing the sample in water at room temperature and allowing the boiling water to cool to room temperature. The sample shall be tested immediately upon removal from the water. Sample size, number of samples, and test method shall be as specified in 4.5.3.1. Samples used for this test shall have been fabricated at least 2 weeks prior to their immersion in the boiling water.

4.5.3.3 Fatigue strength. The fatigue strength shall be determined by subjecting a length of guy with end fittings attached and tensioned to 10 percent of the guy's rated minimum breaking strength to Aeolian (wind induced) vibration a minimum of 2,000,000 cycles. Then, the breaking strength in tension of the specimen shall be determined as specified in 4.5.3.1.

4.5.3.4 Elongation. Elongation shall be determined during the breaking strength in tension test as specified in a through c, unless the testing machine is equipped with an automatic measuring device.

a. Gage marks 762 mm (30 in.) apart shall be made on the sample with a grease pencil.
b. A tape measure, graduated in 1.59 mm (0.062 in.) increments shall be taped to the sample at one gage mark.
c. The other end of the tape shall be attached to a paper clip or similar wire and attached by the clip at the opposite end of the rod and readings shall be made from the tape at the specified loads.

4.5.4 Proof loading tests. Proof loading tests shall be as specified in 4.5.4.1 through 4.5.4.3.

4.5.4.1 Cable elements. Each cable element with end fittings (and insulators, if any) attached shall be proof tested to 55 percent of the minimum breaking strength of the cable. Cables shall be identified with date of proof test and test load. Connections to be made in the field shall be proof load tested during first article test only.

4.5.4.2 Tower climbing device. The tower climbing safety device shall be tested to verify conformance to the load requirements of 3.15.1.2.4.

4.5.4.3 Lifting attachments. Lifting attachments shall be proof loaded to 200 percent of the design load.

4.5.5 Enclosure. Enclosures shall be subjected to the totally enclosed test of MIL-STD-108. This test shall be accomplished on the first article and one from each additional 50 units.

4.5.6 Installability. The antenna shall be installed to verify compliance to the installation instructions, to demonstrate the adequacy of the parts and material provided for installation, and to demonstrate proper ease of installation.

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4.5.6.1 Installability test site. The location of the site for the installability test shall be subject to the approval of the procuring activity. The site selected shall have characteristics which are within the range specified in 3.9.1.15. The site location shall be included in the contractor's test procedures and, once selected, shall not be changed without the approval of the procuring activity.

4.5.7 Weld test. Representative samples of welded structural material shall be subjected to the weld test of AWS D1.1.

4.5.8 Electrical design tests. Unless otherwise specified herein, the tests specified in 4.5.8.1 through 4.5.8.6 shall be conducted on the completely installed antenna subsequent to the installability test of 4.5.6.

4.5.8.1 Impedance and VSWR. Impedance and VSWR measurements shall be made across the entire frequency range of the antenna using swept frequency techniques.

4.5.8.2 DC continuity to ground and contact resistances. The DC continuity to ground and contact resistances shall be demonstrated utilizing a DC resistance meter.

4.5.8.3 Power simulation test. Unless the test requirements of 4.5.8.4 are performed, the contractor shall simulate full power operating conditions on critical components and subassemblies using DC, 60 cycle AC, power sources. Critical components and subassemblies include, but are not limited to, matching unit, input connector feed lines, insulators, and so forth.

4.5.8.4 RF power test. When specified in the contract (see 6.2), full power tests at RF shall be performed at a Government facility. Tests shall be conducted for a minimum of 8 hours per frequency on at least three frequencies. The contractor shall install the antenna as a complete and permanent installation at the Government site. The contractor shall be responsible for all materials and services (for example, site surveys and installation planning) except for primary power and transmitter power, which will be made available by the Government at the transmitter building. The installability test (see 4.5.6) and the remaining electrical design tests may also be performed at the Government site. Frequency assignment will be provided by the Government upon request.

4.5.8.5 Steady-state voltage and frequency. The obstruction lighting system shall be subjected to the steady-state voltage and frequency test of MIL-E-16400.

4.5.8.6 Operating test (obstruction lighting). The obstruction lighting system shall be tested to verify the requirements for control, operation, and intensity.

4.5.9 Thermal design tests. Tests shall be conducted to verify conformance to the requirements of 3.12.

4.5.10 Environmental tests. Environmental tests shall be as specified in 4.5.10.1 through 4.5.10.5.

4.5.10.1 Fungus resistance. Materials used, other than those designated fungus-resistant in Requirement 4 of MIL-STD-454, shall be certified by testing in accordance with Method 508.3 of MIL-STD-810.

4.5.10.2 Humidity test. Electronic assemblies and representative samples of each type of assembled structural materials shall be subjected to the humidity test of MIL-E-16400.

4.5.10.3 Salt fog test. Electronic assemblies and representative samples of each type of assembled structural materials shall be subjected to the salt fog test of MIL-E-16400. All dissimilar metal interfaces shall be included in these samples to evaluate the adequacy of the protection utilized to comply with 3.3.7. Resistance tests shall be performed on all electrical connections before and after the test.

4.5.10.4 Shock test. The obstruction lighting system shall be subjected to the bench handling test of MIL-STD-810, Method 516.3, Procedure VI. The operating test of 4.5.8.6 shall be conducted prior to and after the shock test.

4.5.10.5 EMP. Conformance to the requirements of 3.13.4 shall be verified by test or analysis.

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4.5.11 Safety. Conformance to the requirements of 3.15 shall be verified.

4.5.12 Electromagnetic interference. The obstruction lighting system shall be tested to determine compliance with 3.15.2.1. (A certificate of prior compliance will be acceptable in lieu of testing.)

4.5.13 Leakage current. Conformance to the requirements of 3.15.7 shall be verified.

4.6 Maintenance demonstration. Test procedures of the technical manual and the maintenance standards book shall be verified. The test equipment used to verify the procedures shall be the same as specified in the manuals. Test equipment not specified in the manuals may be used for the manual verification tests only if approved by the procuring activity. This demonstration may be performed during the installability test (see 4.5.6).

4.7 Inspection of preparation for delivery. Inspection shall be performed to ensure conformance with the requirements of Section 5. First article packs which contain Level A packages shall be subjected to rough handling tests as specified in MIL-P-116. Sections packaged for shipment shall be subjected to the rough handling test of MIL-P-116, with the following exception: If the calculated load of the superimposed load test (stackability with dunnage) exceeds 9091 kg (20,000 lbs), the load should be adjusted. The adjusted test load shall be not less than 9091 kg (20,000 lbs).

5. PACKAGING

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in Section 2 do not apply unless specifically stated in the contract. Preparation for delivery requirements for products procured by contractors shall be specified in the individual order.)

5.1 Preservation, packaging, packing, and marking. Unless otherwise specified herein, preparation for delivery shall be in accordance with the applicable levels of preservation, packaging, packing, and marking specified in MIL-E-17555 (see 6.2).

5.1.1 Continuous length nonmetallic guys. Continuous length nonmetallic guys up to 12.70 mm (0.5 in.) diameter maximum may be coiled in manufacture and packaging provided that the coil diameter is not less than 250 times the diameter of the rod.

6. NOTES

6.1 Intended use. The antennas covered by this specification are intended for communications for shore-based Naval Stations at worldwide coastal locations.

6.1.1 Power 10a and Power 50. Power 10a and Power 50 with their high instantaneous values, are intended for use when two 5 kW average, 10 kW peak envelope power (PEP) and two 25 kW average, 50 kW PEP transmitters are respectively multicoupled to one antenna.

6.2 Ordering data. Procurement documents should specify:

a. Title, number, and date of this specification
b. Title, number, and date of the applicable detail specification
c. Power (see 1.2.1)
d. Wind class (see 1.2.2)
e. Obstruction lighting
f. Specific waiver of impedance matching transformer requirements (see 3.7.6)
g. Special grading requirements (see 3.9.1.15)
h. Number of first article samples to be submitted if other than specified in 4.3
i. When reinspected production control and environmental sample antennas may not be accepted (see 4.4.4)
j. RF power test and location, if required (see 4.5.8.4)
k. Levels of preservation, packaging, packing, and marking (see 5.1)

6.3 First article. When a first article is required, it shall be tested and approved under the appropriate provisions of 7-104.55 of the Defense Acquisition Regulation. The first article should be a first production item. The first article should consist of one unit. The contracting officer should include specific instructions in all procurement instruments, regarding arrangements for examinations, tests, and approval of the first article.

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6.4 Definitions. Definitions of terms used in this specification and the applicable detail specification are given in 6.4.1 through 6.4.5.

6.4.1 Structural guy cables. Structural guy cables are cables used to support any pole or tower used in the antenna system.

6.4.2 Cable assemblies other than structural guys. Cable assemblies other than structural guys are all other cables used as radiating elements or cables supporting radiating elements, curtains, or antenna arrays.

6.4.3 Cable elements. Cable elements of the antenna system are structural guy cables and cable assemblies other than structural guys taken collectively.

6.4.4 Gain. Gain is defined as:

$$20 \log_{10} \frac{E_a}{E_i}$$

Where:

E_a = the electric field intensity radiated from the specified antenna over ground in a given direction, with no antenna dissipative or mismatch losses.

E_i = the field intensity which would be radiated from a loss less isotropic source in free space in the same given direction.

Gain as defined herein does not include the effects of antenna efficiency as defined in 6.4.5, but does include ground reflection losses. The existence of farfield conditions and only sky wave radiation is to be assumed. The ground may be either perfectly conducting or imperfectly conducting, but if not otherwise specified in context, perfectly conducting earth may be assumed. If imperfectly conducting earth is specified it shall be assumed to have the properties: $\epsilon_r = 10$; $\sigma = 2$ milliohm per meter.

6.4.5 Efficiency. Antenna efficiency is defined as:

$$E = \frac{\text{Input power to antenna - antenna dissipative losses}}{\text{Input power to antenna}}$$

under the condition of imperfectly conducting ground (that is, $\epsilon_r = 10$; $\sigma = 2$ milliohm per meter). Antenna dissipative losses include ground dissipation (but not ground reflection) losses. Efficiency as defined herein does not include the effect of VSWR.

6.5 Verification procedures. Verification procedures should be as specified in a and b:

a. The technician should install the component of the antenna in the system and adjust as specified in the technical manual.
b. The technician should verify all the procedures of the maintenance standards book and the technical manual.

The technician should have the capabilities and skills of an ET-3 as specified in NAVPERS Manual 18068.

6.6 Special tools. Special tools are defined as those tools not listed in the National Supply Catalog (copies of this catalog may be consulted in the office of the Defense Contract Administration Service).

6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Preparing activity:
NAVY-EC

(Project 5985-N503)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-A-28768B	Antennas, Fixed, High Frequency, General Specification For

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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